

NCI_MEMORY.md

Persistent project memory for OptionScope / NCI Trading. Owner: Michael
(NERDCOMMAND) · GangsterNerds LLC Last updated: 2026-08-18 (replay + exit lab)

How to use this file

Claude does not persist between conversations. Every new session starts blank. This file is the continuity — paste it, or point Claude Code at it, at the start of any OptionScope session.

Update it when a decision is made, not when work is finished. A decision that lives only in a chat log is a decision that gets re-litigated next week.

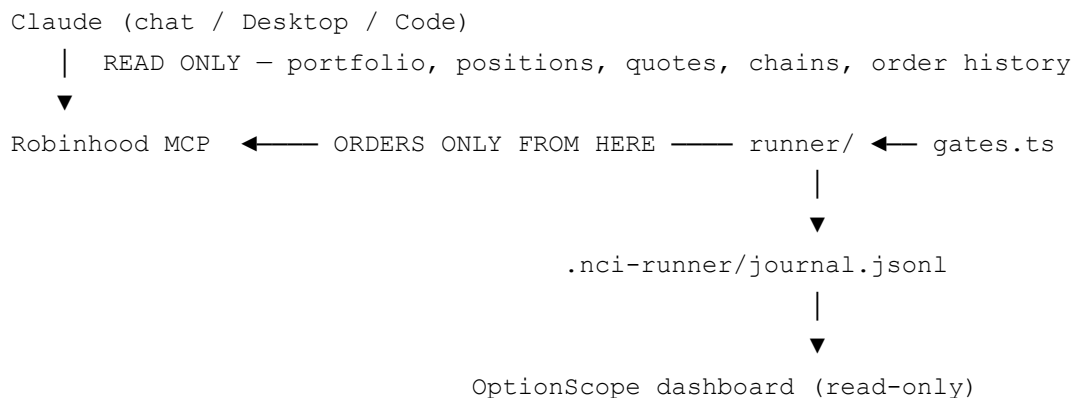
1. What this project is

A systematic options and equities trading platform. Next.js 16 + TypeScript. Repo:
`github.com/chapmancapital1-droid/opiontrading` Local: `D:\GITHUBCLONES\opiontrading`

The trading is a capital engine, not the end goal — it funds NERDCOMMAND Studios and the broader ecosystem.

Scale: ~28,000 lines, 495 files, 197 passing tests across 28 files.

2. Architecture — the rule that governs everything



The rule: the runner is the only thing that places orders.

This is not a preference. If Claude Desktop calls `place_equity_order` directly through the MCP, then `gates.ts` never runs — no \$15 risk cap, no daily halt, no equity floor, no earnings blackout — and nothing is written to the journal, so the dashboard stays empty while real orders fill.

Claude reads. The runner writes. The dashboard displays.

Claude will not place trades. This is a firm limit on Claude’s side, independent of configuration or instruction. Design around it rather than against it — code-enforced risk limits beat model-enforced ones anyway.

3. Current state (2026-08-18)

Robinhood — CONNECTED (read-only)

Connector is live in Claude chat. Five accounts visible.

Account	Type	Options	Agent access
....4849 (default)	Individual · margin	Level 3	No
....3886 “Agentic”	Individual · limited margin	Level 3	YES
....7321	Traditional IRA	Level 2	No
....7739	Roth IRA · cash	—	No
....3844	Joint · margin	—	No

Agentic account3886 — funded and live:

- Total value **\$300.00**, all cash, buying power **\$300.00**
- \$300 showing as pending deposits (still settling)
- Zero open positions, equity or options
- Options level 3 — long calls and puts approved
- `limited_margin` type: can trade unsettled funds, no borrowing

Build status

- `npm run dev` → `localhost:3000/dashboard` — working
- Tabs: **Board** (5 whiteboard panels) and **Robinhood** (4 views)
- Runner: 4 sessions (London scan / pre-market / open / review)
- `runner/robinhood.ts` — read-only MCP client, allowlist-enforced
- Typecheck clean, 197/197 tests, production build compiles

What does NOT exist yet

- No order-placing MCP path in the runner (`paperBroker` only)
- No sync from Robinhood order history into the journal
- No live token for the app itself — the MCP session lives in Claude, not in the Next.js process

4. Guardrails — 4 present, 1 partial, 5 missing

Rendered live in the Robinhood tab → Connection view.

State	Guardrail	Where
PRESENT	Paper/live fail-safe, defaults PAPER	<code>runner/config.ts</code> <code>resolveMode()</code>
PRESENT	Position + loss limits	<code>runner/gates.ts</code> <code>runGates()</code>
PRESENT	Stale-data guard, fails closed	<code>runner/gates.ts</code> <code>PRICES_VALID</code>
PRESENT	Audit logging incl. rejections	<code>runner/state.ts</code> <code>appendJournal()</code>
PARTIAL	Duplicate-order dedup	no time-window on resubmits
MISSING	Order verification via order history	—
MISSING	Ref-ID idempotency on retries	—
MISSING	Kill switch on N consecutive errors	—
MISSING	Human confirmation on options orders	—
MISSING	Atomic multi-leg submission	—

Nothing may call an order tool until the two bolded items land. Without idempotency a retry double-sends; without order-history verification there is no proof a fill happened.

5. Risk configuration

All of it lives in `runner/config.ts`. Nothing else hardcodes a risk value.

Rule	Value
Risk per trade	\$15 fixed dollar (≈ 20 losses to ruin at \$300)
Max concurrent	2
Daily loss halt	\$30
Weekly loss halt	\$60
Equity floor	\$200 — stop and reassess, do NOT add money
Stop loss	Always. ATR-based, capped 5%
Min reward:risk	1.5:1
Max spread	0.6%
Earnings blackout	2 days

Gates **fail closed**. The system may lose money by being wrong about the market. It may never lose money by being confused about its own state.

6. The capital ladder

Hard structural gates, not predictions.

- **\$300 → \$2,000** — Under \$2,000 no margin at all (FINRA 4210(b)(4)). Goal is the 100-trade sample and execution discipline, NOT returns.
- **\$2,000** — Margin minimum cleared. Biggest single unlock. PDT was repealed June 4, 2026, so day trades are unlimited; brokers have until Oct 2027 to implement, so verify Robinhood's status.
- **~\$5,000** — Long options sanely sizable. One ATM contract runs \$150–300; you cannot

sell half a contract, so options need enough capital that one contract is a small position.

- **~\$10,000** — Premium selling / the Wheel. A CSP on a \$20 stock ties up \$2,000 for one contract.
 - **\$25,000+** — The brain's growth modes need this to diversify.
-

7. Validation gate — 100 trades

No live arming until: 100 closed trades, positive expectancy, max drawdown under 20%.
`runner/daily.ts status` reports it and says no with reasons.

Findings so far (simulated, random walk + realistic costs)

93 trades: win rate 44.1%, avg win \$3.81, avg loss \$4.94, expectancy **-\$1.08**, profit factor 0.61, max drawdown 35.3%.

The diagnosis is not the win rate. Wins are smaller than losses because stops fill in full while targets get missed and positions close at the bell. That asymmetry is the thing to fix.

Caveat: the sim is a random walk. There is no trend for signals to find. Losing there proves the strategy isn't free money against noise — a low bar most systems fail — but says nothing about real markets. **Historical replay against actual bars is the test that matters, and it has not been run.**

Bugs the harness caught (all fixed)

1. Stopped clock — RNG seeded per symbol+date, prices never moved
 2. Wrong sequence — day advanced before entry, resolving against a past path
 3. **Halt counters never reset** — would have silently blocked every trade after a bad week, live, with no explanation
-

7b. HISTORICAL REPLAY — real market data (2026-08-18)

Data: 6,570 split-adjusted daily bars pulled via the Robinhood MCP. 10 symbols, 2024-01-02 → 2026-08-14. Cached at `data/bars-daily.json`. Universe deliberately includes losers (NIO 8.42→4.52, SNAP 16.14→5.41) so the test is not survivorship-biased.

Harness: `runner/replay.ts`, `runner/replay-cli.ts` → `REPLAY.bat` No-lookahead by construction — `visibleBars()` withholds future bars rather than trusting signals not to

peek. Gaps below the stop fill at the OPEN. A bar touching both stop and target is assumed to hit the STOP first.

Baseline (fixed 2R exit, 452 trades)

Win rate 46.7% · expectancy **+\$0.40** · profit factor 1.06 · max DD **53.8%** · \$300 → \$480.93

Expectancy went positive on real data. The entry signal is not pure noise. But: **buy-and-hold returned 175.9% (\$300 → \$827.71) over the same window.** 452 trades to underperform a buy button — in a historic bull run for these names. And a 53.8% drawdown is not survivable in practice.

Exit breakdown named the disease: 48.7% timed out (neither stop nor target), only 15.5% reached target, 35.9% stopped including gaps.

Exit lab — 5 plans, 2,058 identical entries (`runner/exit-lab.ts`)

Exit	N	Win%	Expectancy	PF	MaxDD	Final	Resolved
fixed	482	45.4%	\$0.28	1.04	54.9%	\$436	50.4%
breakeven	451	37.7%	\$0.90	1.13	36.8%	\$707	78.5%
trailing	31	25.8%	−\$3.65	0.60	42.6%	\$187	96.8%
ladder	355	51.0%	\$1.66	1.27	22.9%	\$888	45.1%
adaptive	321	53.3%	\$0.86	1.12	40.9%	\$575	86.9%

LADDER wins on every axis that matters — best expectancy, best profit factor, and drawdown cut from 54.9% to 22.9%. It is Michael's own Pair → Pare whiteboard translated to R multiples: scale 50% at 1R, 25% at 2R, run the last 25%. Taking money off the table early is what converts the timeout problem into partial wins.

At minScore 4 the ladder also beats buy-and-hold (\$888 vs \$828).

⚠ THE RESULT IS PARAMETER-FRAGILE — READ THIS BEFORE TRUSTING IT

minScore	ladder expectancy	final equity
4	+\$1.66	\$888
5	−\$2.16	\$196 (hit equity floor)
6	−\$2.73	\$248

Raising the entry filter by ONE point flips the ladder from the best result in the study to a blown account. At minScore 5 every exit plan lands near the \$200 floor — they all stopped out.

A strategy whose sign flips on a one-point parameter change has not demonstrated edge. It has demonstrated a lucky corner of the parameter space. The +\$1.66 number is real arithmetic over real bars, and it is still probably overfit.

Do not fund this on the strength of the minScore-4 row.

What would actually establish edge

- 1. **Walk-forward / out-of-sample split** — fit on 2024, test on 2025–26 unseen
- 2. **Wider universe** — 10 symbols is far too few; 50–100 needed
- 3. **Parameter stability** — the result must survive small perturbations
- 4. **A second window** — ideally one containing a bear market

7c. WALK-FORWARD VALIDATION + EDGE DISCOVERY (2026-08-18)

Data: 30 symbols, 27,210 split-adjusted daily bars, 2023-01 → 2026-08. Includes real disasters (CHPT –97%, LCID –90%, PLUG –81%) — not survivorship-biased.

RESULT 1 — the trend strategy is DEAD

Walk-forward, 3 folds, fit 12mo / test 8mo:

Folds profitable	0 of 3
OOS trades	113
OOS P&L	–\$271.37
In-sample expectancy	+\$4.18/trade
Out-of-sample expectancy	–\$2.40/trade
Edge retained	–57.4%

Sign-flipped, not merely reduced. The optimizer picked different parameters every fold (trailing/5, trailing/5, ladder/4, ladder/5) — never the same twice. That instability is the signature of fitting noise. The +\$1.66 ladder result from the exit lab was curve-fitting. **DO NOT USE** `scoreBullish`.

RESULT 2 — where the edge actually is

Raw conditional-edge scan over 25,080 bars. Base rate: 5-day forward mean +0.430%, win rate 49.0%.

Condition	N	5d edge vs base	Win%
RSI < 25 deep oversold	1,750	+0.48%	54.1%
RSI < 30 oversold	2,989	+0.35%	53.1%
volume 2x + red bar	403	+1.01%	54.1%
gap DOWN >3%	895	+0.33%	50.6%
uptrend (>20MA, 20>50)	7,693	+0.03%	50.7%
pullback RSI<40 in uptrend	859	-0.40%	48.5%
at 60-day high	1,460	-0.15%	23.1%

The edge is MEAN REVERSION — buying fear, not strength. Every condition the old `scoreBullish` voted for sits in the flat-or-negative rows.

Caveat: t-stats reported are against zero, not against the base rate. The signal is supported more by consistency across independent oversold conditions and by the walk-forward result below than by any single t-stat.

RESULT 3 — mean reversion SURVIVED walk-forward

`scoreMeanReversion()` in `runner/sessions.ts`. Same harness, same fills, same discipline that killed the trend strategy:

OOS trades	131
OOS P&L	+\$161.14
In-sample	+\$2.61/trade

Out-of-sample	+\$1.23/trade
Edge retained	47.2%
Verdict	SURVIVED

Honest limits: only 2 folds completed (1 profitable, 1 produced zero trades). 131 OOS trades clears the 100 gate but 2 folds is thin. Needs more folds and a bear-market window before this is fundable.

RESULT 4 — Michael's own 877-trade history says the same thing

From `docs/ota/LESSONS_FROM_YOUR_TRADES.md` :

- Total matched option P/L: **-\$25,264** over 877 trades
- **LONG OPTIONS: 23% win rate, -\$49,305 over 339 trades**
- Assignment events: 107, -\$9,845
- Held to expiration: 130, -\$882
- File's own note: *"Short premium has edge in this sample"*

This is the most important line in the whole project. The Robinhood agentic MCP supports long single-leg options ONLY — which is precisely the category that lost \$49k at a 23% win rate. The agentic path points directly at the historically worst-performing trade type.

Two independent datasets — 27,210 market bars and 877 personal trades — both say the same thing: buying directional strength/premium loses; the edge is elsewhere.

Order of operations, learned the hard way

Find the condition in the data FIRST, then write the strategy around it. The trend strategy was written first and tested afterward, which is how it survived long enough to burn 452 backtested trades.

7d. THE CENTRAL FINDING — Michael has a proven edge (2026-08-18)

Source: `src/knowledge/catalog/ingested/rh_history_lessons.json` , 877 matched option trades. Run `python/edge_analysis.py` to reproduce.

	Trades	Win rate	P&L	Per trade
SHORT options (sell premium)	538	66.9%	+\$24,041	+\$44.69
LONG options (buy premium)	339	23.3%	-\$49,305	-\$145.44
Combined	877	50.1%	-\$25,264	-\$28.81

This is not luck — the sample is large enough to be decisive

- Short win rate 66.9%, **95% CI 62.9%–70.9%** over 538 trades. Entire interval above 50%.
- Long win rate 23.3%, **95% CI 18.8%–27.8%**. Entire interval below 30%.

Both edges are real. One is strongly positive, one is strongly negative, and they were running in the same account.

Had Michael only sold premium and never bought a single option: +\$24,041. The buying side cost \$49,305 and turned a profitable trader into a –\$25k one.

Michael is NOT a bad trader

He is a **profitable premium seller** who also bought directional options. The 50.1% headline win rate hides two completely different strategies. He must not be described, or describe himself, as a bad trader — the data does not support it, and the self-assessment is what makes him vulnerable to overtrading the losing side to “make it back.”

THE CONTRADICTION AT THE HEART OF THE BUILD

The Robinhood agentic MCP places **long single-leg options only** — exactly the 23%-win-rate, –\$49k category. **Michael's proven edge is the one thing the agent cannot place.** Automating the MCP options path would automate his worst strategy at machine speed.

Three honest paths

1. **Equities via the agent** — mean reversion survived walk-forward (+\$1.23 OOS)
2. **Short premium placed manually by Michael**, with OptionScope sizing, gating, and journaling it — his edge, his hands, the app as co-pilot
3. **Wait** for the MCP to support spreads / short legs

Path 2 is the one that uses the demonstrated edge. It also happens to be the one where the app's real job is risk management and position sizing, not signal generation — which is where the personal history says the failure was (avg win \$125 vs avg loss \$183, 130 held to

expiration, 107 assignments).

The payoff-ratio problem, stated plainly

- Personal trades: avg win \$125.40, avg loss \$183.37 → PF 0.685
 - Today's backtest: avg win \$3.81, avg loss \$4.94 **The same asymmetry appears in the human record and in the code.** It is structural, not emotional. The ladder exit cut backtest drawdown 54.9% → 22.9%. Fixing the payoff ratio is worth more than improving entry accuracy.
-

8. Decisions made and why

- **London 3–6am ET scans, does not execute.** London is a forex session. For US equities that window is pre-market — thin book, wide spreads. The scan stages ideas; 9:35 trades them into real liquidity. Override: `sessions.london.allowExecution`.
 - **Stops are never optional.** `STOP_PRESENT` is a hard gate.
 - **Fixed dollar sizing, not percentage.** At \$300, percentage sizing produces units the spread eats.
 - **Vercel cancelled.** Ephemeral filesystem would destroy the journal. Local until other people use it, then a real database.
 - **Journal is append-only.** A trade log you can quietly edit will eventually flatter you.
 - **"Aggressive but high probability" is a contradiction.** Opposite ends of one dial. Pick a point; you cannot have both ends.
 - **Doubling \$300 in five months = ~15%/month compounded with no losing month.** Several times the best hedge fund in history, on a small account, while learning. Not a low-probability plan — a near-zero one.
 - **Do not trade to fund a subscription.** Emotional weight beyond the dollars destroys position sizing. Claude does not need funding and does not persist.
-

9. Options education baked into the app

Board tab (5 panels) and Robinhood → Education.

The four trades. Buy = you get a right and pay for it. Sell = you take a duty and get paid for it. Call = up matters. Put = down matters. Every contract is 100 shares; premium is quoted

per share.

Buyers can only lose what they paid. Sellers can lose far more than they collected. The agent path starts on the buy side — and the MCP only supports long single-leg orders today anyway.

Eleven strategies, four postures: Directional (buy/sell call/put), Bi-directional (straddle), Protective (protective put, variation, collar, buy/hold), Non-directional (credit spread, iron condor).

Delta ladder: 0.8–1.0 deep ITM (moves like shares) · 0.5–0.8 ITM · ~0.5 ATM (coin flip, most time value, fastest theta) · 0–0.3 OTM (cheap lottery, where most beginner losses live).

Pair → Pare exit ladder: enter 4, pare 2 at +150%, pare 1 at +100%, run the last on a trailing stop, –50% hard stop on the whole position. The panel computes whether the two pares actually recover entry capital before the runner is left open.

Volatility: price orbits value; the distance is emotion. Compressed IV favors buyers, expanded favors sellers.

10. Next actions, in order

1. ~~Ref ID idempotency~~ — DONE, `runner/idempotency.ts`, 15 tests
2. ~~Order history verification~~ — DONE, `runner/verify.ts`
3. ~~Historical replay~~ — DONE, `runner/replay.ts` + real bar cache
4. ~~Exit logic~~ — DONE, `runner/exits.ts`. Ladder wins; result is fragile.
5. ~~Walk forward validation~~ — DONE. Trend strategy INVALIDATED (–\$2.40 OOS). Mean reversion SURVIVED (+\$1.23 OOS).
6. **More folds + a bear-market window** for mean reversion — only 2 folds completed. THE blocking item now.
7. **Resolve the long-options contradiction** — personal history says long options lose 23%; the MCP only supports long options. Either the equity path is the product, or options need spreads the MCP cannot place.
8. Kill switch on N consecutive errors
9. Human confirmation on options orders
10. Only then: the gated order path

11. Working agreement

- Honest pushback on framing and risk is part of the job, not friction
- Show the failing number, don't bury it
- Verify before asserting — Claude wrongly claimed twice that no Robinhood connector existed, and was corrected only by actually calling the tool. **Call the tool. Don't describe what you assume it would return.**
- Michael sets direction; Claude reads the codebase and builds
- Michael is not "a loser" and not a bad trader. He is a Navy veteran running a company who shipped a 28,000-line trading platform, AND a demonstrably profitable premium seller (538 trades, 66.9%, +\$24k). Broke is a bank balance, not a verdict. When he calls himself a bad trader, show him the `by_side` table — the data refutes it.